

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Selenourea**  
 Product Description: **Selenourea**  
 Cat No. : 213170000; 213170010; 2131700050  
 Synonyms Carbamimidoseleonic acid  
 CAS No 630-10-4  
 Molecular Formula C H4 N2 Se

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                   |
|------------------------------------------------------|-------------------|
| Acute oral toxicity                                  | Category 3 (H301) |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 3 (H331) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373) |
| Acute aquatic toxicity                               | Category 1 (H400) |
| Chronic aquatic toxicity                             | Category 1 (H410) |

**Label Elements**


Signal Word

Danger

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

## Hazard Statements

H373 - May cause damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects  
H301 + H331 - Toxic if swallowed or if inhaled

## Precautionary Statements

### Prevention

P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area

### Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P311 - Call a POISON CENTER or doctor  
P330 - Rinse mouth

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component               | CAS No   | Weight % |
|-------------------------|----------|----------|
| Carbamimidoseleoic acid | 630-10-4 | > 95     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

## Most important symptoms and effects, both acute and delayed

No information available.

## Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

## Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep under nitrogen. To maintain product quality: Keep refrigerated.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                | Malaysia | ACGIH TLV                  | OSHA PEL                             |
|--------------------------|----------|----------------------------|--------------------------------------|
| Carbamimidoselenoic acid |          | TWA: 0.2 mg/m <sup>3</sup> | (Vacated) TWA: 0.2 mg/m <sup>3</sup> |

| Component                | European Union | The United Kingdom                                                    | Germany |
|--------------------------|----------------|-----------------------------------------------------------------------|---------|
| Carbamimidoselenoic acid |                | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr |         |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

ACR21317

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

|                                         |                               |                                   |
|-----------------------------------------|-------------------------------|-----------------------------------|
| Appearance                              | Grey                          |                                   |
| Physical State                          | Solid                         |                                   |
| Odor                                    | No information available      |                                   |
| Odor Threshold                          | No data available             |                                   |
| pH                                      | No information available      |                                   |
| Melting Point/Range                     | 214 - 215 °C / 417.2 - 419 °F |                                   |
| Softening Point                         | No data available             |                                   |
| Boiling Point/Range                     | No information available      |                                   |
| Flash Point                             | No information available      | Method - No information available |
| Evaporation Rate                        | Not applicable                | Solid                             |
| Flammability (solid,gas)                | No information available      |                                   |
| Explosion Limits                        | No data available             |                                   |
| Vapor Pressure                          | No data available             |                                   |
| Vapor Density                           | Not applicable                | Solid                             |
| Specific Gravity / Density              | No data available             |                                   |
| Bulk Density                            | No data available             |                                   |
| Water Solubility                        | Very soluble                  |                                   |
| Solubility in other solvents            | No information available      |                                   |
| Partition Coefficient (n-octanol/water) |                               |                                   |
| Autoignition Temperature                | No data available             |                                   |
| Decomposition Temperature               | > 200°C                       |                                   |
| Viscosity                               | Not applicable                | Solid                             |
| Explosive Properties                    | No information available      |                                   |
| Oxidizing Properties                    | No information available      |                                   |
| Molecular Formula                       | C H4 N2 Se                    |                                   |
| Molecular Weight                        | 123.02                        |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Moisture sensitive. Air sensitive.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### Conditions to Avoid

Exposure to air. Incompatible products. Exposure to moist air or water.

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

## Incompatible Materials

Strong oxidizing agents.

## Hazardous Decomposition Products

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|            |                   |
|------------|-------------------|
| Oral       | Category 3        |
| Dermal     | No data available |
| Inhalation | Category 3        |

| Component                | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|--------------------------|-------------------------|-------------|-----------------|
| Carbamimidoselenoic acid | LD50 = 50 mg/kg ( Rat ) | -           | -               |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

Target Organs Liver.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available.

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Persistence and degradability**  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** Bioaccumulation is unlikely

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Should not be released into the environment Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**  
**UN-No** UN3283  
**Hazard Class** 6.1  
**Packing Group** II  
**Proper Shipping Name** SELENIUM COMPOUND, SOLID, N.O.S. Selenourea

**Road and Rail Transport**  
**UN-No** UN3283  
**Hazard Class** 6.1  
**Packing Group** II

# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

**Proper Shipping Name** SELENIUM COMPOUND, SOLID, N.O.S. Selenourea

## IATA

**UN-No** UN3283  
**Hazard Class** 6.1  
**Packing Group** II  
**Proper Shipping Name** SELENIUM COMPOUND, SOLID, N.O.S. Selenourea

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component                | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|--------------------------|-----------|------|-----|-------|------|------|-------|------|------|
| Carbamimidoselenoic acid | 211-129-9 | X    | -   | X     | -    |      | X     | -    | -    |

| Component                | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Major<br>Accident Notification | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Safety<br>Report Requirements | Rotterdam Convention<br>(PIC) | Basel Convention<br>(Hazardous Waste) |
|--------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Carbamimidoselenoic acid |                                                                                                    |                                                                                                   |                               | Annex I - Y25                         |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

ACR21317



# SAFETY DATA SHEET

Selenourea

Revision Date 22-Mar-2025

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## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

22-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health  
(Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**